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ICE 3.2



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1. (3 marks) It is known that the time complexity of a brute force attack on a 2-stage DES is not significantly higher than it is for a single-stage DES. Does this mean that a 2-stage DES reduces to a 1-stage DES?

- Yes, since time-complexity of a brute-force attack on a 2-stage DES and a 1-stage DES are about the same
- **No, if that were the case then a 3-stage DES can be reduced to a 2-stage DES which can then be reduced to 1-stage DES (correct)**
- It is unclear what is meant by "2-stage DES reduced to 1-stage DES"
- None of the above

2. (2 marks) What is meant by "Avalanche effect" in DES crypto-systems?

- "Avalanche effect" implies that a change in any one bit in plaintext will change every alternate bit in the ciphertext
- **This is the same as "diffusion", where a change in any one bit of plaintext changes approximately half the number of ciphertext bits (correct)**
- "Avalanche effect" implies that a change in any one bit in plaintext will change 32 bits in the ciphertext
- None of the above

3. (3 marks) Consider DES. Part of its description, viz. the $F(.)$ function in any given DES round, is elaborated below, together with the "table" that describes the "Straight P-box" therein. "P" is short for "permutation". What is the 4-th bit in the output of the Permutation box in relation to the input bits to the Permutation box? And as what output bit does 16-th bit in the input show up? (Kindly refer the form for the P-box):

- The 4-th output bit is the 31-st input bit, and 16-th input bit shows up as the 10-th bit in the output
- **The 4-th output bit is the 21-st input bit, and 16-th input bit shows up as the 1-st bit in the output (correct)**
- The 4-th output bit is the 31-st input bit, and 16-th input bit shows up as the 1-st bit in the output
- The 4-th output bit is the 21-st input bit, and 16-th input bit shows up as the 10-th bit in the output



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